

Lesson 5.3: Utility: overall preference

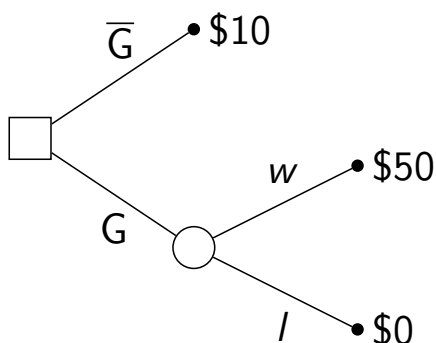
Risk attitudes and utility

Victor Jauregui

Subjective elements: beliefs and preferences

Example (Sports betting)

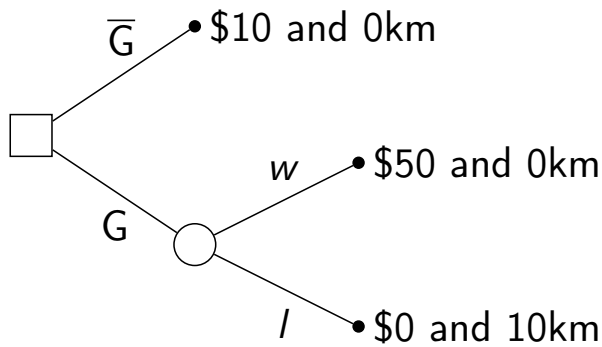
After an evening out, Alice has \$10 left to pay for her 10km trip home. Her favourite sports team is about to play one of the league favourites, so she stops to check the odds. For a \$10 bet, she can get \$50.



Alice's beliefs and preferences

- ▶ believes her team's chances of winning are two in five:
 $P(w) = \frac{2}{5}$
- ▶ would bet only if $P(w) \geq \frac{1}{2}$

Monetary payoff vs outcome preference



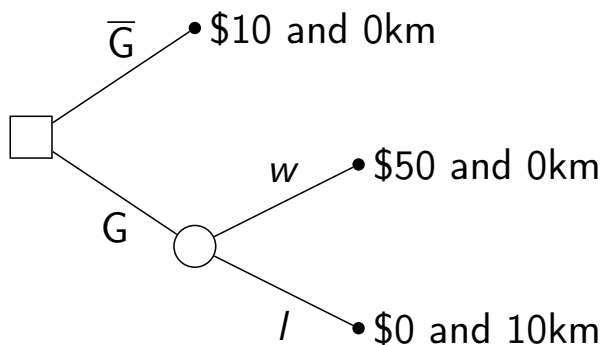
$$\begin{aligned}v(\bar{G}) &= v(\$10) \\ &= 10\end{aligned}$$

$$\begin{aligned}v(G) &= v\left(\left[\frac{2}{5} : \$50 \mid \frac{3}{5} : \$0\right]\right) \\ &= \frac{2}{5}v(\$50) + \frac{3}{5}v(\$0) \\ &= 20\end{aligned}$$

- ▶ assumed that $v(\$x) = x$
- ▶ ignored preference for (not) walking

Utility: overall outcome preference

- ▶ Utility: $u(\$x)$ is overall preference value of outcome with monetary payoff $\$x$
- ▶ Alice's certainty equivalent for $\left[\frac{1}{2} : \$50 \mid \frac{1}{2} : \$0\right]$ is \$10



$$u(\bar{G}) = u(\$10)$$

$$\begin{aligned}u(G) &= u\left(\left[\frac{1}{2} : \$50 \mid \frac{1}{2} : \$0\right]\right) \\ &= \frac{1}{2}u(\$50) + \frac{1}{2}u(\$0)\end{aligned}$$